



Predict With Events

Guess both landings, then run to check

Name:

Date:

★ I can predict where both sprites land by following the events that start them.

Predict the whole chain

Good coders predict before they run. The green flag starts Bit, and Bit's last block broadcasts 'go'. Pixel starts 'when I receive go'. To predict where BOTH land, follow the events: green flag → Bit moves and sends 'go' → Pixel hears 'go' and moves. Predict both, then run to check.

The number path



Bit

Bit's script

when green flag clicked

forward 3

broadcast 'go'



Pixel

Pixel's script

when I receive 'go'

back 2

1 Predict — before you run

Follow the events in your head. PREDICT where each lands. Bit (home 1, forward 3) will land on: ____

Pixel (home 5, back 2) will land on: ____

2 Run and check

Now run it: click the flag, let Bit broadcast 'go', then run Pixel. Where did each really land? Bit: ____

Pixel: _____. Did your prediction match?

3 What started Pixel?

Pixel had no green-flag block. What event started Pixel's script? Write it.

Big idea

Predicting with events means following the chain: which event starts the first script, what message it sends, and which script that message starts next. Then each sprite's moves tell you where it lands.



Predict With Events

Quick facilitation notes & answer key

What this worksheet teaches

Students PREDICT the whole event chain before running: green flag starts Bit, Bit moves and sends a message, the message starts Pixel. A correct prediction names BOTH landings. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: two counters on a number line 0 to 5 (green Bit, purple Pixel), and a card that says "go" you can hand over to act out sending a message.

How to lead it (about 20 minutes)

1. Show them (you do). Follow the chain in your head: flag → Bit moves and sends "go" → Pixel runs.
2. Do one together. Exercise 1 — predict Bit on 4 and Pixel on 3.
3. Let them try. Students run to check (Ex 2), then name what started Pixel (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 (predict) — Bit: 1, forward 3 → 4 (predict 4). Pixel: 5, back 2 → 3 (predict 3).

Exercise 2 (run) — the run confirms Bit on 4 and Pixel on 3.

Exercise 3 — Pixel was started by the message "go" (its "when I receive go" hat). The green flag only directly starts Bit; Pixel depends on Bit's message.

If students get stuck — and ways to adjust

Watch for: predicting only Bit. The chain ends with Pixel, so predict both.

Make it easier: predict Bit, follow the message, then predict Pixel.

Make it harder: ask them to predict whether the two meet.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions that are started by an event, like clicking the green flag or getting a message.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change what starts them or one move, and explain what the change did.

You'll know it worked when

The child predicts both landings by following the event chain, then checks by running.